Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $a, b, c \neq 0$ and $a+b+c=0$, then the matrix $\begin{bmatrix} 1+\frac{1}{a} & 1 & 1 \\ 1 & 1+\frac{1}{b} & 1 \\ 1 & 1 & 1+\frac{1}{c} \end{bmatrix}$ is

(A) singular

(B) non-singular

(C) skew-symmetric

(D) orthogonal

Q.2

If B is an idempotent matrix, and $A = I - B$, then

(A) $A^2 = A$

(B) $A^2 = I$

(C) $\mathrm{AB} = \mathrm{I}$

(D) $\mathrm{BA} = \mathrm{I}$

Q.3

If A and B are matrices of same order, then $\left(\mathrm{AB}' - \mathrm{BA}'\right)$ is a

(A) skew symmetric matrix

(B) null matrix

(C) symmetric matrix

(D) unit matrix

Q.4

If A is symmetric as well as skew-symmetric matrix, then A is

(A) Diagonal

(B) Null

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(C) Triangular

(D) None of these

Q.5

If C is skew-symmetric matrix of order n and X is $\mathrm{n} \times 1$ column matrix, then $\mathrm{X}'\mathrm{CX}$ is a

(A) scalar matrix

(B) unit matrix

(C) null matrix

(D) None of these

Q.6

If $\mathrm{A}_1,\ \mathrm{A}_3, \ldots \ldots, \mathrm{A}_{2\mathrm{n}-1}$ are n skew-symmetric matrices of same order,
then $\mathrm{B}=\sum_{\mathrm{r}=1}^{\mathrm{n}}(2\mathrm{r}-1)\left(\mathrm{A}_{2\mathrm{r}-1}\right)^{2\mathrm{r}-1}$ will be

(A) symmetric

(B) skew-symmetric

(C) neither symmetric nor skew-symmetric

(D) data is adequate

Q.7

Elements of a matrix A of order 10×10 are defined as $a_{ij}=w^{i+j}$ (where w is cube root of unity), then $\mathrm{tr}(\mathrm{A})$ of the matrix is

(A) 0

(B) 1

(C) 3

(D) None of these

Q.8

If A and B be two square matrices of order λ whose all the elements are essentially positive integers then the minimum value of tr $\left(\mathrm{AB}^2\right)$ is equal to

(A) λ^3

(B) λ^2

(C) $2\lambda^2$

(D) none of these

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Q.9

The element a_{ij} of square matrix is given by $a_{ij} = (i+j)(i-j)$, then matrix A must be

(A) Skew-symmetric matrix

(B) Triangular matrix

(C) Symmetricmatrix

(D) Null matrix

Q.10

Let A and B be 3×3 matrices of real numbers, where A is symmetric, B is skew symmetric, and $(A+B)(A-B) = (A-B)(A+B)$. If $(AB)^t = (-1)^k AB$ where $(AB)^t$ is the transpose of the matrix AB, then k is

(A) any integer

(B) odd integer

(C) even integer

(D) cannot say anything

Q.11

If A is a square matrix, then AA^T is a

(A) skew-symmetric matrix

(B) symmetric matrix

(C) diagonal matrix

(D) None of these

Q.12

If matrix $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ then find $\text{tr}(A) + \text{tr}\left(A^2\right) + \text{tr}\left(A^3\right) + \ldots + \text{tr}\left(A^{100}\right)$

(A) 100

(B) 50

(C) 200

(D) None of these

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Q.13

The values of a, b, c if $\begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is orthogonal

(A) $\mathrm{a} = \pm\frac{1}{\sqrt{2}}; \mathrm{b} = \pm\frac{1}{\sqrt{6}}; \mathrm{c} = \pm\frac{1}{\sqrt{3}}$

(B) $\mathrm{a} = \pm\frac{1}{\sqrt{2}}; \mathrm{b} = \pm\frac{1}{\sqrt{3}}; \mathrm{c} = \pm\frac{1}{\sqrt{6}}$

(C) $\mathrm{a} = \pm\frac{1}{\sqrt{6}}; \mathrm{b} = \pm\frac{1}{\sqrt{2}}; \mathrm{c} = \pm\frac{1}{\sqrt{3}}$

(D) $\mathrm{a} = \pm\frac{1}{\sqrt{3}}; \mathrm{b} = \pm\frac{1}{\sqrt{2}}; \mathrm{c} = \pm\frac{1}{\sqrt{6}}$

Q.14

Let $\mathrm{A} + 2\,\mathrm{B} = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and

$2\,\mathrm{A} - \mathrm{B} = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$, then $\mathrm{tr}(\mathrm{A}) - \mathrm{tr}(\mathrm{B})$ is

(A) 1

(B) 3

(C) 2

(D) 0

Q.15

If A and B are symmetric matrices of the same order and $\mathrm{X} = \mathrm{AB} + \mathrm{BA}$ and $\mathrm{Y} = \mathrm{AB} - \mathrm{BA}$, then $(\mathrm{XY})^{\mathrm{T}}$ is equal to

(A) XY

(B) YX

(C) -YX

(D) None of these

Q.16

If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$ then value of α for which $\mathrm{A}^2 = \mathrm{B}$, is

(A) 1

(B) -1

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(C) 4

(D) No real values

Q.17

Which of the following is/are correct?

(A) $\mathrm{B}'\mathrm{AB}$ is symmetric if A is symmetric

(B) $\mathrm{B}'\mathrm{AB}$ is skew-symmetric if A is symmetric

(C) $\mathrm{B}'\mathrm{AB}$ is symmetric if A is skew-symmetric

(D) None of these

Q.18

If Z is an idempotent matrix, then $(\mathrm{I}+\mathrm{Z})^n$

(A) $\mathrm{I}+2^n\mathrm{Z}$

(B) $\mathrm{I}+\left(2^n-1\right)\mathrm{Z}$

(C) $\mathrm{I}-\left(2^n-1\right)\mathrm{Z}$

(D) None of these

Q.19

If A and B be symmetric matrices of the same order, then $AB-BA$ will be a

(A) Symmetric matrix

(B) Skew symmetric matrix

(C) Null matrix

(D) None of these

Q.20

If $A=\begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 5 \\ 2 & -5 & 0 \end{bmatrix}$, then

(A) $A'=A$

(B) $A'=-A$

(C) $A'=2A$

(D) None of these

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Q.1 Answer:

non-singular

Solution:

$$\begin{vmatrix} 1+1/a & 1 & 1 \\ 1 & 1+1/b & 1 \\ 1 & 1 & 1+1/c \end{vmatrix}$$

$$= \frac{1}{abc}\begin{vmatrix} a+1 & b & c \\ a & b+1 & c \\ a & b & c+1 \end{vmatrix}$$

Using $C_1 \to C_1 + C_2 + C_3$**, we get** $|A| = \frac{1}{abc}(a+b+c+1)\begin{vmatrix} 1 & b & c \\ 1 & b+1 & c \\ 1 & b & c+1 \end{vmatrix}$

Using $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$**, we get**

$$|A| = \frac{1}{abc}(a+b+c+1)\begin{vmatrix} 1 & b & c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= \frac{1}{abc}(a+b+c+1) \neq 0$$

Q.2 Answer:

$A^2 = A$

Solution:

Since B **is an idempotent matrix,** $\therefore B^2 = B$**. Now,** $A^2 = (I-B)^2 = (I-B)(I-B)$

$= \mathrm{I} - \mathrm{IB} - \mathrm{BI} + \mathrm{B}^2 = \mathrm{I} - \mathrm{B} - \mathrm{B} + \mathrm{B}^2 = \mathrm{I} - 2\,\mathrm{B} + \mathrm{B}^2$

$= \mathrm{I} - 2\,\mathrm{B} + \mathrm{B} = \mathrm{I} - \mathrm{B} = \mathrm{A}$

$\therefore$ A is idempotent.

Q.3 Answer:

skew symmetric matrix

Solution:

$$\left(\mathrm{AB}' - \mathrm{BA}'\right)' = \left(\mathrm{AB}'\right)' - \left(\mathrm{BA}'\right)'$$

$$= \left(\mathrm{B}'\right)'\mathrm{A}' - \left(\mathrm{A}'\right)'\mathrm{B}' = \mathrm{BA}' - \mathrm{AB}' = -\left(\mathrm{AB}' - \mathrm{BA}'\right)$$

Hence, $\left(\mathrm{AB}' - \mathrm{BA}'\right)$ **is a skew-symmetric matrix.**

Q.4 Answer:

Null

Solution:

Let $\mathrm{A} = [\mathrm{a_{ij}}]_{\mathrm{n}\times\mathrm{m}}$**. Since A is skew-symmetric** $\mathrm{a_{ii}} = 0$ $(\mathrm{i} = 1, 2, \ldots\ldots, \mathrm{n})$ **and** $\mathrm{a_{ji}} = -\mathrm{a_{ji}} (\mathrm{i} \neq \mathrm{j})$

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Also, A is symmetric so $a_{ji} = a_{ji} \forall i$ and $j \therefore \quad a_{ji} = 0 \quad \forall \quad i \neq j$
Hence $a_{ij} = 0 \forall i$ and $j \Rightarrow A$ is a null zero matrix

Q.5 Answer:

null matrix

Solution:

Here, X is $n \times 1$, C is $n \times n$ and X' is $1 \times n$ Hence, $\mathrm{X'CX}$ is 1×1 matrix. Let $\mathrm{X'CX = k}$.
Then, $(X'CX)' = X'C'X' = X'C'X' = X'(-C)X$
$= -\mathrm{X'CX} = -\mathrm{k}$
$\mathrm{k} = -\mathrm{k} \Rightarrow \mathrm{k} = 0 \Rightarrow \mathrm{X'CX}$ **is null matrix.**

Q.6 Answer:

skew-symmetric

Solution:

$$\mathrm{B} = \mathrm{A}_1 + 3\ \mathrm{A}_3^3 + \ldots..(2\mathrm{n}-1)\left(\mathrm{A}_{2\mathrm{n}-1}\right)^{2\mathrm{n}-1}$$
$$\mathrm{B}^{\mathrm{T}} = -\left[\mathrm{A}_1 + 3\ \mathrm{A}_3^3 + \ldots.(2\mathrm{n}-1)\left(\mathrm{A}_{2\mathrm{r}-1}\right)^{2\mathrm{r}-1}\right]$$
$$= -\mathrm{B},\ \text{so skew-symmetric}$$

Q.7 Answer:

None of these

Solution:

$\mathrm{tr(A)} = \sum_{\mathrm{i=j}} \mathrm{a_{ij}} = \left(\mathrm{a_{11}} + \mathrm{a_{22}} + \mathrm{a_{33}} + \ldots + \mathrm{a_{10}}\right)$
$= \left(\mathrm{w}^2 + \mathrm{w}^4 + \mathrm{w}^6 + \ldots + \mathrm{w}^{20}\right) = \mathrm{w}^2\left(1 + \mathrm{w}^2 + \mathrm{w}^4 + \ldots + \mathrm{w}^{18}\right)$
$= w^2\left[\left(1 + w + w^2\right) + \ldots + \left(1 + w + w^2\right) + 1\right] = w^2 \times 1 = w^2$

Q.8 Answer:

λ^2

Solution:

$\mathrm{tr(A)} = \sum_{\mathrm{i}=1}^{\lambda} \mathrm{a_{ii}} \geq \lambda \quad (\ \textbf{As}\ \mathrm{a_{ij}} \geq 1 + \mathrm{i,j})$
and $\mathrm{tr}\left(\mathrm{MN}^2\right) = \{\mathrm{tr(M)}\}\{\mathrm{tr(N)}\}^2$

Q.9 Answer:

Skew-symmetric matrix

Solution:

We have $a_{\mathrm{ji}} = (j + \mathrm{i})(\mathrm{j} - \mathrm{i}) = -(\mathrm{i} + \mathrm{j})(\mathrm{i} - \mathrm{j}) = -\mathrm{a}$
Hence, matrix is skew-symmetric.

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Q.10 Answer:

odd integer

Solution:

$$(\mathrm{A}+\mathrm{B})(\mathrm{A}-\mathrm{B})=(\mathrm{A}-\mathrm{B})(\mathrm{A}+\mathrm{B})$$
$$\Rightarrow \quad \mathrm{AB}=\mathrm{BA}$$

as A is symmetric and B is skew-symmetric $(\mathrm{AB})^{\mathrm{t}}=-\mathrm{AB}$

$\Rightarrow \mathrm{k}$ **is an odd integer.**

Q.11 Answer:

symmetric matrix

Solution:

$$\text{Let } \mathrm{A}=\begin{bmatrix}1 & -1 & 1\\ 2 & 1 & 0\\ 1 & -1 & 2\end{bmatrix}, \text{ then}$$
$$\mathrm{A}'=\begin{bmatrix}1 & 2 & 1\\ -1 & 1 & -1\\ 1 & 0 & 2\end{bmatrix}$$
$$\mathrm{AA}'=\begin{bmatrix}1 & -1 & 1\\ 2 & 1 & 0\\ 1 & -1 & 2\end{bmatrix}\begin{bmatrix}1 & 2 & 1\\ -1 & 1 & -1\\ 1 & 0 & 2\end{bmatrix}=\begin{bmatrix}3 & 1 & 4\\ 1 & 5 & 1\\ 4 & 1 & 4\end{bmatrix}$$

Q.12 Answer:

100

Solution:

Consider

$$\mathrm{A}^2=\begin{bmatrix}-5 & -8 & 0\\ 3 & 5 & 0\\ 1 & 2 & -1\end{bmatrix}\begin{bmatrix}-5 & -8 & 0\\ 3 & 5 & 0\\ 1 & 2 & -1\end{bmatrix}$$
$$=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=\mathrm{I}\ \textbf{So}\ \mathrm{A}^3=\begin{bmatrix}-5 & -8 & 0\\ 3 & 5 & 0\\ 1 & 2 & -1\end{bmatrix}\ \textbf{and}$$

so on $\mathrm{tr}+\mathrm{tr}(\mathrm{A})\left(A^2\right)+\mathrm{tr}\left(A^3\right)+\ldots+\mathrm{tr}\left(A^{100}\right)$

$=(-1)+(3)+(-1)+(3)+\ldots+(-1)+(3)=150-50=100$

Q.13 Answer:

$\mathrm{a}=\pm\frac{1}{\sqrt{2}};\mathrm{b}=\pm\frac{1}{\sqrt{6}};\mathrm{c}=\pm\frac{1}{\sqrt{3}}$

Solution:

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Let $A = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$

Now, $\mathrm{A}^{\mathrm{T}} = \begin{bmatrix} 0 & \mathrm{a} & \mathrm{a} \\ 2\,\mathrm{b} & \mathrm{b} & -\mathrm{b} \\ \mathrm{c} & -\mathrm{c} & \mathrm{c} \end{bmatrix}$

$\because$ A is orthogonal $\therefore \mathrm{AA}^{\mathrm{T}} = \mathrm{I}$

$$\Rightarrow \begin{bmatrix} 0 & 2\,\mathrm{b} & \mathrm{c} \\ \mathrm{a} & \mathrm{b} & -\mathrm{c} \\ \mathrm{a} & -\mathrm{b} & \mathrm{c} \end{bmatrix} \begin{bmatrix} 0 & \mathrm{a} & \mathrm{a} \\ 2\,\mathrm{b} & \mathrm{b} & -\mathrm{b} \\ \mathrm{c} & -\mathrm{c} & \mathrm{c} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Equating the corresponding elements, we get $4\,\mathrm{b}^2 + \mathrm{c}^2 = 1 \quad \ldots$ (1)

$2\,\mathrm{b}^2 - \mathrm{c}^2 = 0$ ...(2)

$a^2 + b^2 + c^2 = 1$ ...(3)

On solving $(1), (2)$ and (3), we get $\mathrm{a} = \pm\frac{1}{\sqrt{2}}; \mathrm{b} = \pm\frac{1}{\sqrt{6}}; \mathrm{c} = \pm\frac{1}{\sqrt{3}}$

Q.14 Answer:

2

Solution:

Here to find the value of $\mathrm{tr}(\mathrm{A}) - \mathrm{tr}(\mathrm{B})$, we need not to find the matrices A and B. We can find tr $(\mathrm{A}) - \mathrm{tr}(\mathrm{B})$ using the properties of trace of matrix, ie., $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix} \Rightarrow \mathrm{tr}(A + 2B) = -1$ ...(1)

or $\mathrm{tr}(\mathrm{A}) + 2\,\mathrm{tr}(\mathrm{B}) = -1$

$$\Rightarrow \quad 2\,\mathrm{A} - \mathrm{B} = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$$

$\Rightarrow \mathrm{tr}(2\,\mathrm{A} - \mathrm{B}) = 3$ or $2\,\mathrm{tr}(\mathrm{A}) - \mathrm{tr}(\mathrm{B}) = 3(2)$ ...(2)

Solving (1) and (2), we get $\mathrm{tr}(\mathrm{A}) = 1$ and $\mathrm{tr}(\mathrm{B}) = -1 \Rightarrow \mathrm{tr}(\mathrm{A}) - \mathrm{tr}(\mathrm{B}) = 2$

Q.15 Answer:

-YX

Solution:

Given that $\mathrm{X} = \mathrm{AB} + \mathrm{BA}$

$\Rightarrow X = X^{\mathrm{T}}$ and $Y = AB - BA \quad \Rightarrow \quad Y = -Y^T$

Now $(XY)^{\mathrm{T}} = Y^{\mathrm{T}}X^{\mathrm{T}} = -YX$

Q.16 Answer:

No real values

Solution:

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$$A^2 = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix}$$

Clearly, no real value of a.

Q.17 Answer:

$\mathrm{B}'\mathrm{AB}$ is symmetric if A is symmetric

Solution:

Let A be a symmetric matrix. Then, $\mathrm{A}' = \mathrm{A}$

Now, $(\mathrm{B}'\mathrm{AB})' = \mathrm{B}'\mathrm{A}'(\mathrm{B}')' \cdot [\because (\mathrm{AB})' = \mathrm{B}'\mathrm{A}']$

$= \mathrm{B}'\mathrm{A}'\mathrm{B} \quad [\because (\mathrm{B})' = \mathrm{B}]$

$= \mathrm{B}'\mathrm{AB} \quad [\because \mathrm{A}' = \mathrm{A}]$

$\Rightarrow \mathrm{B}'\mathrm{AB}$ **is a symmetric matrix.**

Now, let A be a skew-symmetric matrix. Then, $\mathrm{A}' = -\mathrm{A}$

$\therefore \quad (\mathrm{B}'\mathrm{AB})' = \mathrm{B}'\mathrm{A}'(\mathrm{B}')' \quad [\because (\mathrm{AB})' = \mathrm{B}'\mathrm{A}']$

$= \mathrm{B}'\mathrm{A}'\mathrm{B} \quad [\because (\mathrm{B}')' = \mathrm{B}]$

$= \mathrm{B}'(-\mathrm{A})\mathrm{B} \, [\because \mathrm{A}' = -\mathrm{A}]$

$= -\mathrm{B}'\mathrm{AB} \quad \therefore \mathrm{B}'\mathrm{AB}$ **is a skew-symmetric matrix.**

Q.18 Answer:

$\mathrm{I} + (2^n - 1)\,\mathrm{Z}$

Solution:

Z **is idempotent then** $\mathrm{Z}^2 = \mathrm{Z} \Rightarrow \mathrm{Z}^3, \mathrm{Z}^4, \ldots$

$\mathrm{Z}^n = \mathrm{Z}$

$(\mathrm{I} + \mathrm{Z})^n = {}^n\mathrm{C}_0\mathrm{I}^n + {}^n\mathrm{C}_1\mathrm{I}^{n-1}\mathrm{Z} + {}^n\mathrm{C}_2\mathrm{I}^{n-2}\mathrm{Z}^2 + \ldots + {}^n\mathrm{C}_n\mathrm{Z}^n$

$= {}^nC_0 I + {}^nC_1 Z + {}^nC_2 Z + {}^nC_3 Z + \ldots + {}^nC_n Z$

$= \mathrm{I} + ({}^n\mathrm{C}_1 + {}^nC_2 + {}^n\mathrm{C}_3 + \ldots + {}^n\mathrm{C}_n)\,\mathrm{Z} = \mathrm{I} + (2^n - 1)\,\mathrm{Z}$

Q.19 Answer:

Skew symmetric matrix

Solution:

Since A, B **are symmetric** $\Rightarrow A = A'$ **and** $B = B' \therefore (AB - BA)' = (AB)' - (BA)' = B'A' - A'B'$

$= -(A'B' - B'A') = -(AB - BA) \Rightarrow (AB - BA)$ **is skew-symmetric.**

Q.20 Answer:

$A' = -A$

Solution:

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$$A' = \begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & -5 \\ -2 & 5 & 0 \end{bmatrix} = -A$$

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